

Dyslexia Assistance System

Date: 1/31/26

Programme: A Dyslexia Assistance System that empowers people with print disabilities and enhances their reading experience

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I. Background and Problem Statement:

1.1 Project background

Dyslexia is a common neurodevelopmental disorder characterized by difficulties with literacy, reading, and spelling that cannot be explained by intellectual, educational, or sensory problems.^[1] This is due to the brain's congenital disorders in the conversion of "glyphs→phonetics→semantics", rather than personal intelligence or learning attitudes.

The disorder occurs in all age groups worldwide and has a wide impact. In learning and working, people with dyslexia face significant challenges: students are inefficient in reading textbooks and completing homework; Adults have difficulty understanding when dealing with literature and work documents, which affects the quality of learning and work efficiency.^[2]

In the digital age, most textual content is not tailored to the needs of the dyslexic population. Global research on dyslexia assistance is still in the early stages, leading to a notable gap in the development of professional and targeted auxiliary tools for this group.

1.2 Core questions

The core issues faced by the current group of individuals with reading difficulties can be summarized into the following four categories^[1]:

(1) Difficulty in character recognition: Easily confusing characters with similar shapes (such as ‘b’ and ‘d’, ‘土’ and ‘士’), or reversing the order of words and sentences, leading to obstacles in basic reading^[3].

(2) Low reading fluency: Needing to read word by word, unable to achieve "whole word recognition," resulting in slow reading speed, frequent rereading, and easily distracted attention, making it hard to focus on text comprehension.

(3) Obstacles derived from spelling and writing: Difficulty in recognizing characters directly leads to a high rate of spelling errors, while a weak understanding of the phonetic and semantic associations of complex vocabulary affects information absorption.

(4) Accompanied by cognitive challenges: Some patients have weak working memory, easily distracted attention, and may experience psychological stress and a sense of avoidance when facing large blocks of text.

1.3 Limitations of existing solutions

There are already some reading assistance tools available in the market, such as Helperbird^[4], Microsoft Immersive Reader^[5], Kurzweil 3000^[6], etc., but these tools generally have the following shortcomings:

(1) Single-function focus: Most tools only focus on visual optimization (such as adjusting font size, background color), lacking processing of the text content itself (such as splitting long sentences, replacing complex vocabulary), making it difficult to fundamentally reduce the semantic understanding barrier for readers with reading difficulties;

(2) Limited scenario coverage: Existing tools are mostly suitable for general reading scenarios, lacking targeted support for specific scenarios such as academic literature, student assignments, etc., and some tools lack multimodal support (such as synchronized voice following and text highlighting);

(3) Low level of personalization: Different types of reading difficulties (surface-level, deep-level, phonetic) were not considered, making it difficult to provide customized support solutions.

1.4 Project Significance

This project aims to develop a multimodal adaptive reading assistance system for individuals with dyslexia, addressing the core reading pain points of those with dyslexia from a technical perspective. The implementation of this project holds three key significances:

(1) Social significance: Focus on the information access rights of people with dyslexia, a vulnerable group, narrow the digital divide, and promote inclusive social development

(2) Academic significance: Combining technologies such as NLP and large language models with the cognitive psychology characteristics of dyslexia, exploring innovative development models for digital assistive tools, providing practical references for research in related fields;^[7]

(3) Practical significance: Providing efficient, convenient, and personalized reading assistance solutions for individuals with dyslexia, covering multiple scenarios such as student learning and adult work, enhancing their learning and work efficiency, and strengthening their sense of social participation.^[8]

II.Objectives and Outcome

2.1 Project objectives

This project will develop a dyslexia-assisted reading system that integrates text optimization, visual adaptation, and multi-modal assistance to solve the core problems of dyslexia in glyph recognition, reading fluency, semantic understanding, etc., and improve their reading experience and efficiency. The specific objectives are as follows:

(1) Realize the intelligent optimization function of text^[9]: support long sentence splitting, short sentence merging, and complex vocabulary replacement to ensure that the length of a single sentence conforms to the cognitive habits of dyslexics (no more than 15-20 words in Chinese and no more than 12 words in English).

(2) Provide personalized visual adaptation solutions: built-in dyslexia-friendly fonts, support word spacing, line spacing, and font size adjustment, and provide high-contrast color matching schemes to reduce the burden of glyph recognition;

Develop multimodal auxiliary tools: implement functions such as text-to-speech, keyword annotation, and vocabulary phonetic example query to meet the learning habits of people with different dyslexias.

(3) Adapt to multiple scenarios and languages: Support the import of common document formats such as PDF, adapt to mainstream languages such as Chinese and English, and cover scenarios such as academic reading and daily learning.

(4) Optimize the system interaction experience: adopt a minimalist interface design to reduce interference elements, and the operation steps are no more than 3 steps, lowering the threshold for user use.

2.2 Expected Outcome

2.2.1 Product results

Runnable auxiliary reading system prototype (including front-end interface and back-end service), supporting web access;

System function document: Detail the functional modules, operation process, technical parameters, etc. of the system;

User Manual: A simple operation guide for people with print disabilities, including feature introductions, configuration suggestions, and more.

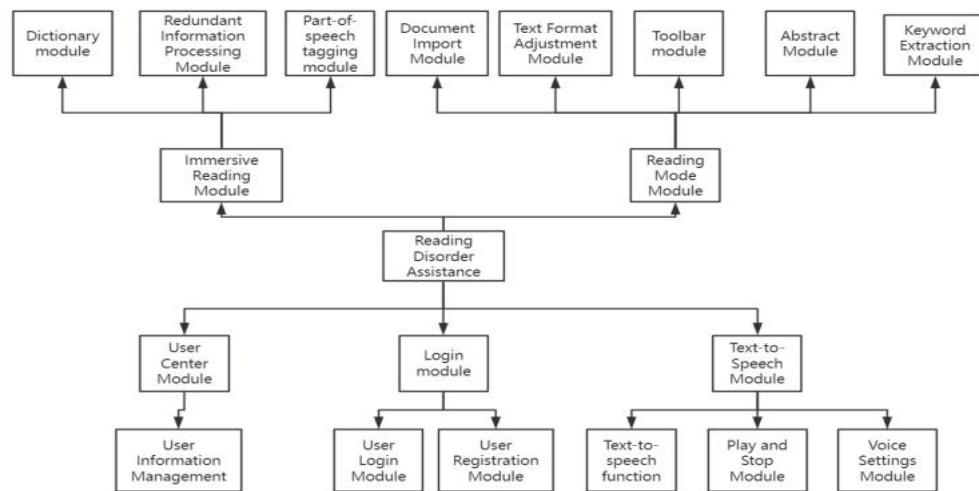
2.2.2 Technological achievements

- (1) **Text optimization algorithm model:** long sentence splitting and vocabulary replacement algorithm developed based on spaCy, HanLP and other tools;
- (2) **Multimodal data interaction interface:** realize efficient interaction of text, voice, visual configuration and other data;
- (3) **System source code and open source documentation:** Ensure the reusability and scalability of the project.

2.3 Project Contributions

- (1) **Technological innovation:** Integrate NLP and large language model technology to achieve intelligent optimization of text content, breaking through the limitations of existing tools that only focus on visual adjustments.
- (2) **Literacy Adaptability:** Conduct special optimization for the reading and writing needs of dyslexic individuals, and fit their core literacy usage scenarios.
- (3) **Practical value:** Provide a "one-stop" reading aid solution that takes into account personalization and ease of use, and can be directly applied to the learning and work of people with dyslexia;

Overview of the top-level requirements of the system's figure as follow:



Picture 1 - top-level requirements of the system

III. Project Methodology

3.1 Literature Review

Dyslexia is a typical neurodevelopmental disorder, the core cause is abnormal brain information processing, which has nothing to do with IQ^[1], visual deficits or acquired

education, and its core pain points are concentrated in four aspects^[2]: difficulty in recognizing glyphs due to confusing similar words and reversing word order, low reading fluency caused by spelling and frequent re-reading, spelling and writing disorders caused by high misspelling rate, chaotic strokes and accompanying cognitive challenges, which are embodied in weak working memory and easy distraction.

Existing studies have identified effective intervention paths, that is, improving the reading experience through visual presentation optimization, text structure simplification, and multimodal support, such as OpenDyslexic^[10] special fonts to reduce font confusion, and Bionic Reading keyword annotation to improve reading fluency. However, the current market products have obvious limitations: Helperbird^[4], Microsoft Immersive Reader^[5], etc., focus on visual optimization (font, color matching, spacing adjustment) and ignore the in-depth processing of text content; Although Google Simplify uses large language models to simplify text, it has a single function; Most products lack personalized adaptation and special optimization for Chinese context, and Kurzweil 3000^[6] still has the problems of complex operation and high price.

Domestic research focuses on the diagnosis of dyslexia and traditional teaching intervention, and the development of digital auxiliary tools lags behind. Foreign product technology is mature but insufficient adaptability. NLP tools such as spaCy^[11] and HanLP^[12], FastAPI^[13] and Spring Boot frameworks^[14], and the development of large language models provide technical support for the integration of "vision + structure + semantics" three-dimensional optimization, which can realize intelligent text simplification, user type recognition and dynamic adaptation, and lay the foundation for the development of Chinese-friendly, comprehensive and personalized reading assistance systems.

3.2 Project stage division

We have divided the project into 5 phases:

- (1) **Requirements analysis and literature research stage (weeks 1-2):** clarify user needs, complete literature review and technical research;
- (2) **Technology Solution Design Phase (Weeks 3-4):** Determine system architecture, functional modules, technology stack, and tool selection;

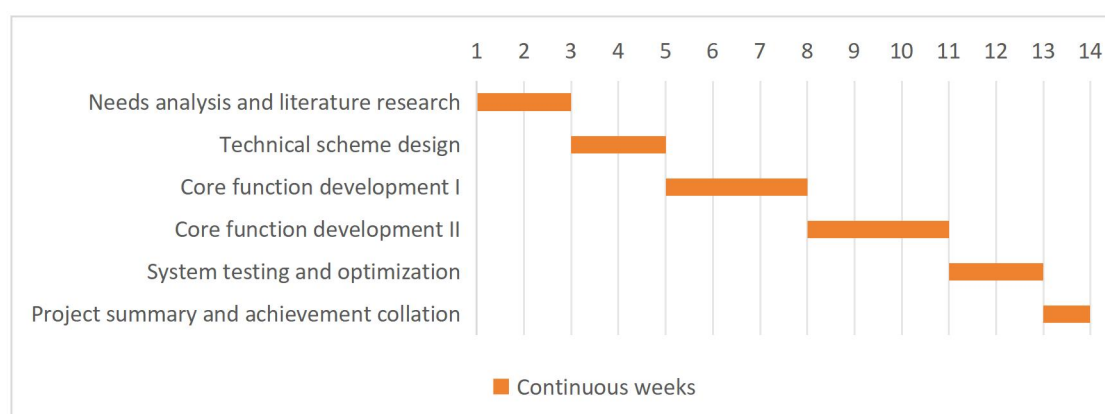
(3) Core feature development phase (weeks 5-10): Implement core functions such as text optimization, visual adaptation, and module multimodal assistance;

(4) System Testing and Optimization Phase (Weeks 11-12): Conduct functional testing, performance testing, and user experience testing, and optimize the system based on feedback.

(5) Project Summary and Results Summary Phase (Week 13): Full project report, user manual, demonstration video, and other outcomes.

IV. Project Schedule

Here is the Gantt chart of this project:



Picture 2 - System's Gantt chart

(1) The tasks of weeks 1-2 are to complete the literature research related to dyslexia, clarify the list of user needs and functions, and write a literature review, with the milestone being to submit a literature review and needs analysis report.

(2) The tasks of Weeks 3-4 are to determine the system architecture and technology stack, design the database table structure, draw the front-end interface prototype, and the milestone is to submit the technical solution design document and UI prototype.

(3) The tasks of weeks 5-7 are to implement back-end user management and data storage interfaces, develop Python text optimization algorithms, and complete basic communication between the front and back ends.

(4) The tasks for weeks 8-10 are to realize the front-end visual adaptation function (font, color matching, spacing adjustment), develop the voice reading and vocabulary query function, and complete the front-end and back-end joint debugging.

(5) The tasks of Weeks 11-12 are to conduct functional and performance tests, invite people with dyslexia to conduct user experience testing, and optimize system vulnerabilities and interactive experience based on feedback, with milestones such as submitting system test

reports and optimized versions.

(6) **The tasks of Week 13** are to write the final project report and user manual, record system demonstration videos, organize the source code and related documentation, and the milestones are to submit the full project deliverables and complete the project defense.

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